CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

1st Semester of M. Pharm (Quality Assurance)

University Theory Examination December 2016

PHQA001 MODERN ANALYTICAL TECHNIQUES

Date: 12.12.16 , Monday **Time:10:00 a.m to 01:00 p.m** **Maximum Marks: 70**

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1 **A)** Explain in brief about Derivative UV Spectroscopy with its application. **05**

B) Enlist the ionization techniques used in mass spectroscopy. Discuss MALDI techniques. **05**

Q.2 **A)** Explain following terms with suitable examples: **05**

Auxochromes, Bathochromic shift, Spin-Spin Splitting, Q-Absorption ratio and transition in UV Spectroscopy.

B) Give the structure, Name of compound and Explain each value on basis of following details. **05**

Mol,wt:73 gm/mole, IR:3413 cm^{-1},3260 cm^{-1},2900 cm^{-1},1667 cm^{-1},1460 cm^{-1}, NMR: 6.5 δ (13sq, singlet), 2.25 δ (12.8sq, quartet),1.10 δ (19.7sq, triplet).

OR

Q.2 **A)** Define Chemical Shift. Explain Factor affecting Chemical shift. **05**

B) Give the structure, Name of compound and Explain each value on basis of following details. **05**

Mol. wt:69 gm/mole, IR:2941 cm^{-1},2273 cm^{-1} ,1460 cm^{-1}, UV: no absorbance above 200 nm, NMR:2.72 δ (4.2 sq, septet), 1.34 δ (25 sq, doublet).

Q.3 **Answer the following (Any three)** **15**

(A) Give principle of IR spectroscopy. What is FT-IR.

(B) Enlist mass analyzers. Explain Time of Flight (TOF) in detail.

(C) Write a Principle and Application of NMR Specroscopy.

(D) Explain Woodward-Fischer rules for calculating absorbance.

SECTION-II

Q.4	(A)	Enlist detectors used in Gas chromatography? Explain them.	05
	(B)	What are components of HPLC? Write on detail about photo diode array.	05
Q.5	(A)	Define the term Retention time, Resolution, Capacity factor, Distribution Constant and Theoretical plate.	05
	(B)	Write a note on Ion exchange Chromatography.	05

OR

Q.5	(A)	Write a note on Differential Thermal Analysis.	05
	(B)	Derive Bragg's Law and write on X ray diffraction.	05
Q.6		**Answer the following (Any three)**	15
	(A)	Describe ELISA technique and application of Radio Immuno assay?	
	(B)	What is Flourescence and Phosphorescence? Factors affecting quenching.	
	(C)	Write a note on instrumentation of differential scanning colorimetry (DSC).	
	(D)	Write a note on Capillary Electrophorosis.	